Final Report: Lillian Hatch

Background/Motivation for the Study

Mental health challenges are widespread and found in people of all ages. Researchers have identified specific lifestyle interventions that positively affect mental health symptoms in people with both acute and chronic difficulties. It is commonly held that physical activity is positively correlated with certain mental health disorders (Mahindru, 2023). Similarly, physical activity has been shown to enhance sleep quality, which can be highly deregulated among those suffering from mental health conditions (Fuligni, 2018). According to Mahindru et al., "increased physical activity can help attenuate some psychotic symptoms and treat medical comorbidities that accompany psychotic disorders." When access to mental health care is limited for many individuals, lifestyle interventions like physical activity and sleep are highly accessible and modifiable. These are promising methods of managing mental health issues for people regardless of their ability to receive other types of care.

Research Question and Hypothesis

This project attempts to answer how the frequency of physical activity and sleep duration predict poor mental health days over 30 days. This question can be broken down into two components.

First, based on existing literature, it is reasonable to expect a relationship between physical activity frequency and the number of poor mental health days. Beyond this, I anticipate a minimum threshold of physically active days where benefits become noticeable. The number of poor mental health days will have an inverse relationship with the number of days of physical activity. A maximum threshold could emerge, beyond which the relationship is less pronounced. For this component, at its most basic level, the null hypothesis is that there is no relationship between the frequency of physical activity and the number of poor mental health days. The alternative hypothesis is that the frequency of physical activity will have a negative relationship with the number of poor mental health days.

Second, there may be a correlation between median sleep duration and the number of days with poor mental health. Because it seems that there is an ideal range for optimal sleep duration, and both hypersomnia and insomnia are associated with specific mental health challenges, it is hypothesized that sleep duration that falls outside the optimal range will be correlated with poorer mental health days. For this component, the null hypothesis is that there is no relationship between sleep duration and the number of poor mental health days. The alternative hypothesis is that the sleep duration will also have an inverse relationship, or even a U-shaped relationship, with the number of poor mental health days.

Data Description and Exploratory Data Analysis

Read in the NHANES dataset, filtering for the targeted variables relevant to the research question. For this analysis, the DaysMentHlthBad, PhysActiveDays, and SleepHrsNight variables are included.

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

The NHANES dataset consists of health-related information captured by survey responses from study participants. DaysMentHlthBad is the number of days of bad mental health in 30 days. PhysActiveDays reflects the number of physically active days in a week. SleepHrsNight is the self-reported hours of sleep per night.

```
[1] 3523

 DaysMentHlthBad  PhysActiveDays  SleepHrsNight
 Min.   : 0.000   Min.   :1.000   Min.   : 2.000
 1st Qu.: 0.000   1st Qu.:2.000   1st Qu.: 6.000
 Median : 0.000   Median :3.000   Median : 7.000
 Mean   : 3.875   Mean   :3.705   Mean   : 6.971
 3rd Qu.: 4.000   3rd Qu.:5.000   3rd Qu.: 8.000
 Max.   :30.000   Max.   :7.000   Max.   :12.000

[1] 7.454153

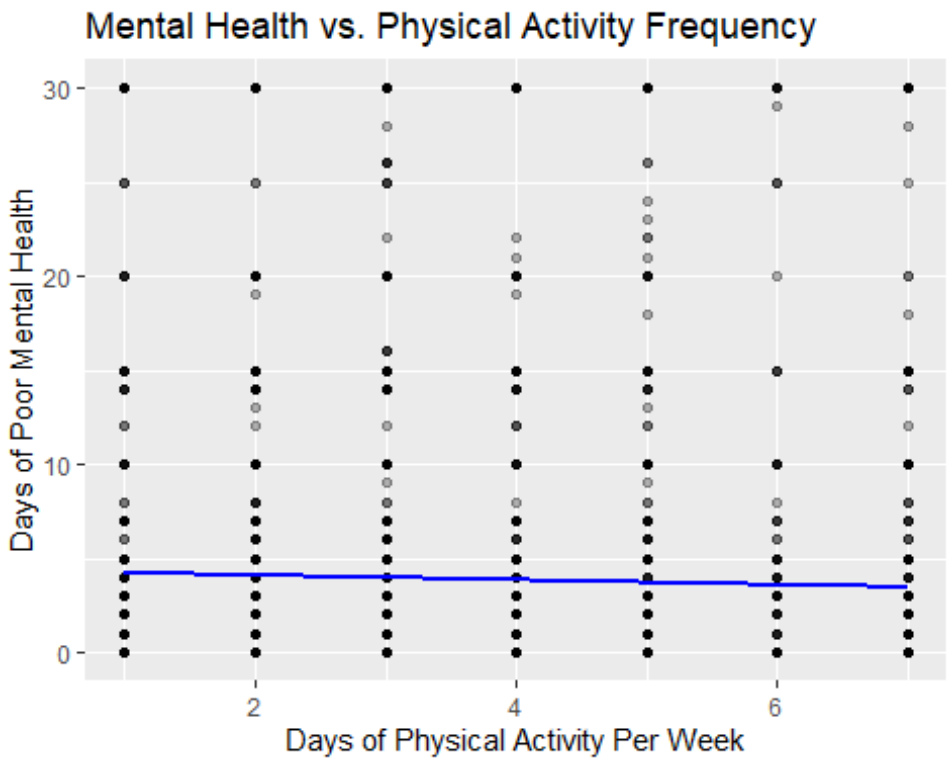
[1] 1.81421

[1] 1.311733
```

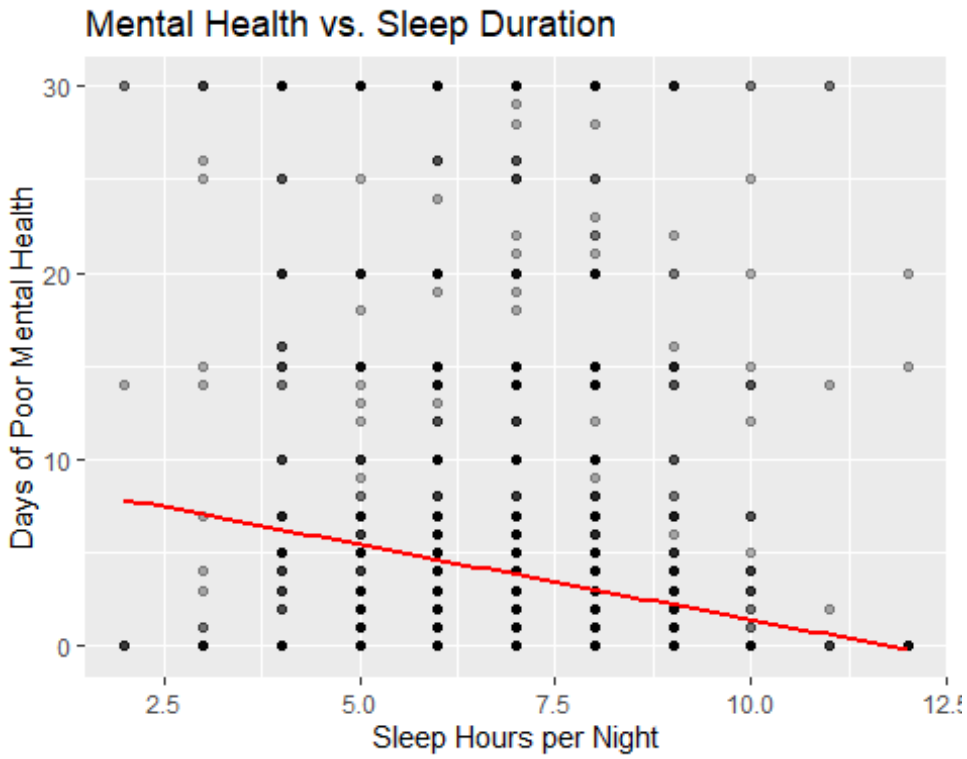
The original dataset has a sample size of 10,000 individuals, but after filtering out missing values the final sample size for this research question is 3,523 individuals. DaysMentHlthBad ranges from 0 to 30 days, with a mean of 3.88 days and a median of 0. This indicates a right-skewed distribution where most participants report few poor mental health days. Additionally, the standard deviation is 7.45, which shows wide variability in the number of mental health days reported. PhysActiveDays ranges from 1 to 7, with a mean of 3.71 and a median of 3. This suggests moderate physical activity levels across the sample. The standard deviation is 1.81. SleepHrsNight ranges from 2-12 hours, with a mean of 6.97, a median of 7 and a standard deviation of 1.31. This shows that most participants report sleep duration near the recommended range.

Scatterplots are used for both visualizations because they allow for clear representation of relationships between days of poor mental health and the frequency of physical activity, as well as days of poor mental health and duration of sleep. Scatterplots are ideal here because they allow for easy detection of trends and correlations.

```
`geom_smooth()` using formula = 'y ~ x'
```




```
`geom_smooth()` using formula = 'y ~ x'
```



There is a slight inverse relationship between days of physical activity and days of poor mental health. This warrants additional analysis as clearly some relationship is displayed, but indicates that frequency of physical activity may only be a small factor in the number of poor mental health days per month. The scatterplot visualizing days of poor mental health and sleep duration shows a slightly stronger inverse relationship. Participants who sleep more tend to report fewer days with poor mental health. This shows a distinct connection between sleep and mental health, though it does not support the initial hypothesis. This dataset reveals no optimal range of sleep for mental health, as initially expected, just an inverse relationship. Most noteworthy from these graphs is that mental health outcomes vary widely, even among those with similar sleep durations and those with similar physical activity frequency.

Analysis

Linear regression is used to evaluate the relationships between the frequency of physical activity and mental health, and sleep duration and mental health. This method will allow us to quantify the strength and significance of these associations. The first model evaluates only physical activity frequency and mental health. The second model evaluates only sleep duration and mental health. And the last model combines these two predictors to show how each behaves while controlling for the other.

```
Call:
lm(formula = DaysMentHlthBad ~ PhysActiveDays, data = nhanes_filtered)
```

```
Residuals:
    Min      1Q  Median      3Q     Max
-4.2155 -3.9637 -3.4599  0.0363 26.5401

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     4.34148    0.28553   15.21   <2e-16 ***
PhysActiveDays -0.12594    0.06921   -1.82   0.0689 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.452 on 3521 degrees of freedom
Multiple R-squared:  0.0009395, Adjusted R-squared:  0.0006558
F-statistic: 3.311 on 1 and 3521 DF,  p-value: 0.06889


Call:
lm(formula = DaysMentHlthBad ~ SleepHrsNight, data = nhanes_filtered)

Residuals:
    Min      1Q  Median      3Q     Max
-7.8753 -3.8518 -3.0470  0.1482 29.3671

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    9.4847     0.6725  14.104   <2e-16 ***
SleepHrsNight  -0.8047     0.0948  -8.488   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.38 on 3521 degrees of freedom
Multiple R-squared:  0.02005,   Adjusted R-squared:  0.01977
F-statistic: 72.05 on 1 and 3521 DF,  p-value: < 2.2e-16


Call:
lm(formula = DaysMentHlthBad ~ PhysActiveDays + SleepHrsNight,
    data = nhanes_filtered)

Residuals:
    Min      1Q  Median      3Q     Max
-7.8282 -3.9342 -2.8985  0.1468 29.7416

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     9.89973    0.71506  13.845   <2e-16 ***
PhysActiveDays -0.11679    0.06854  -1.704   0.0885 .
SleepHrsNight  -0.80216    0.09479  -8.463   <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.378 on 3520 degrees of freedom
Multiple R-squared:  0.02086,   Adjusted R-squared:  0.0203
F-statistic:  37.5 on 2 and 3520 DF,  p-value: < 2.2e-16
```

The first model showed only a small negative relationship between physical activity and poor mental health days, with an estimated slope of -0.13. This means that for each additional day of physical activity in a week there are 0.13 fewer poor mental health days per month on average. Though there is a relationship, it is very small, and the p-value, p=0.07, shows it is not statistically significant. The r-squared value, R^2=0.0009, shows that the model explains less than 0.1% of the variation in mental health outcomes.

The second model showed sleep duration was a statistically significant predictor of the number of poor mental health days. The p-value was <0.0001, and the estimated slope (-0.80) shows that for each additional hour of sleep there are 0.80 fewer poor mental health days on average. The r-squared value was 0.020, which means that the model explains about 2% of the variation in mental health outcomes.

When both predictors were taken together sleep remained statistically significant, $p < 0.0001$, controlling for frequency of physical activity. Physical activity was still not significant when controlling for sleep duration, with a p-value of 0.089. The combined model showed only slightly more variance than the sleep-only model (R^2=0.021), though this is minimal and doesn't meaningfully improve the model.

Conclusions

From the linear regressions, we found that although there is a negative relationship between the frequency of physical activity and the number of days of poor mental health per month, it is not statistically significant. The null hypothesis cannot be rejected; therefore, based on this analysis and data, it cannot be concluded that increasing physical activity in a week decreases the number of poor mental health days. The hypothesis for this component of the analysis was incorrect, and there is no meaningful relationship, let alone a peak number of physical activity days, where the mental health benefits are most noticeable.

Sleep duration, on the other hand, is strongly correlated with a decrease in the number of poor mental health days. The null hypothesis is rejected, indicating a statistically significant relationship between the two. The initial hypothesis was incorrect because a U-shaped relationship was predicted; instead, a negative linear relationship exists.

The combined model reinforces this data. Once again, the frequency of physical activity is not statistically significant, but the duration of sleep remains significant. The R^2 for each model was low, which indicates that these factors account for only a small portion of the variation in mental health outcomes. This aligns with the initial scatterplots, which showed a wide scatter in poor mental health days across all levels of physical activity and sleep.

Sleep duration appears to account for only a small portion of the variation in mental health. There are many other factors at play.

From this project, we can see that there is a relationship between the amount of sleep a person gets each night and the number of poor mental health days per month. Physical activity doesn't appear to have a statistically significant effect on poor mental health days. Even though there is clearly more to mental health than getting a good night's sleep, a full night's rest is clearly beneficial.

References

Fuligni, A. J., Arruda, E. H., Krull, J. L., & Gonzales, N. A. (2018). Adolescent Sleep Duration, Variability, and Peak Levels of Achievement and Mental Health. *Child development*, *89*(2), e18–e28. https://doi.org/10.1111/cdev.12729

Mahindru, A., Patil, P., & Agrawal, V. (2023). Role of Physical Activity on Mental Health and Well-Being: A Review. *Cureus*, *15*(1), e33475. https://doi.org/10.7759/cureus.33475